

COOPERATIVE BEHAVIOR IN BIOLOGICAL SYSTEMS

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Fröhlich's model of Bose condensation in biological systems is discussed. A criticism of the theory levied by Lifshits is examined utilizing microscopic techniques. It is subsequently concluded that the criticism is invalid.

Cooperative behavior in biological systems. The possible participation of the vibrations of the dipolar segments of biological macromolecules in fundamental biological processes has been discussed repeatedly [1–10]. Fröhlich [1–5] proposed that such coherent behavior is feasible if energy is supplied to the dipolar constituents above a certain critical rate which causes the oscillating subunits to condense into the lowest energy mode in a manner characteristic of Bose condensation. Fröhlich's analysis was based upon rate equations whereby each dipolar segment was considered to be suspended in a heat bath.

Subsequently, other investigators [6–10] approached the theoretical problem from differing standpoints. Lifshits [6] concluded that the Fröhlich model was erroneous because omitted in the rate equations is a "two quantum" term which, when included, obliterates the Bose condensation effect.

The purpose of the present paper is to illustrate that Lifshits' argument is incorrect, and that Bose condensation does occur even when the Lifshits term is included in the calculation.

According to Fröhlich, the biological system is presumed to consist of (i) oscillating units of giant dipoles (e.g., sections of the membrane of cells or parts of H-bonds within biological macromolecules), (ii) the remainder of the biological system which formed the heat bath, and (iii) an external energy source which couples to the vibratory units. Long-range Coulomb interaction between the oscillatory units produces a narrow band of frequencies, $\omega_i(\omega_0 \leq \omega_i \leq \omega_{\max})$, corresponding to the normal modes of the electromagnetic oscillations. Utilizing these assumptions, Fröhlich deduced that the average rate of change of i th mode quanta is [2]

$$\dot{\langle n_i \rangle} = S - \phi[\langle n_i \rangle \exp(\hbar\omega_i/kT) - (1 + \langle n_i \rangle)] - \chi \sum_j [\langle n_i \rangle (1 + \langle n_j \rangle) \exp[\hbar(\omega_i - \omega_j)/kT] - \langle n_j \rangle (1 + \langle n_i \rangle)], \quad (1)$$

where S is the rate of energy supplied, and ϕ and χ are coupling constants to the heat bath for the one- and two-quantum processes respectively.

Lifshits claims that included in the interaction hamiltonian should be "two-quantum" terms which allow the simultaneous creation and destruction of the quanta (i) and (j). Thus, according to Lifshits, a modification of the Fröhlich formula is required which yields for the average rate of change of i th mode quanta [6]:

$$\dot{\langle n_i \rangle} = S - \phi[\langle n_i \rangle \exp(\hbar\omega_i/kT) - (1 + \langle n_i \rangle)] - \sum_j \chi_j [\langle n_i \rangle (1 + \langle n_j \rangle) \exp(\hbar\omega_i/kT) \quad (2)$$

$$- \langle n_j \rangle (1 + \langle n_i \rangle) \exp(\hbar\omega_j/kT) - (1 + \langle n_i \rangle)(1 + \langle n_j \rangle) + \langle n_i \rangle \langle n_j \rangle \exp[\hbar(\omega_i + \omega_j)/kT]]$$

$$= S - B_i[\langle n_i \rangle \exp(\hbar\omega_i/kT) - (1 + \langle n_i \rangle)], \quad (3)$$

where $\chi_j = \chi \exp(-\hbar\omega_j/kT)$, $B_i = \sum_j \chi_j [1 + \langle n_j \rangle + \langle n_j \rangle \exp(\hbar\omega_j/kT)] + \phi$. Therefore, the two-quantum term is of the same form as the one-quantum term and no Bose condensation occurs.

To confirm the weakness of Lifshits' conclusions, the problem must be treated microscopically. Assigning creation and destruction operators to each mode associated with the oscillating subunits (a_i^+ , a_i), heat bath (b_k^+ , b_k), and the energy source (p_p^+ , p_p), it is possible to write the hamiltonian of the biological system as:

$$H = \sum_i \hbar\omega_i a_i^+ a_i + \sum_k \hbar\Omega_k b_k^+ b_k + \sum_p \hbar\bar{\Omega}_p p_p^+ p_p + \sum_{i,j,k} (\lambda a_i b_j^+ b_k + \lambda^* a_i^+ b_j b_k^+) \\ + \sum_{i,j,k} (\chi a_j^+ a_i b_k^+ + \chi^* a_j a_i^+ b_k) + \frac{1}{2} \sum_{i,j,k} (\beta a_j a_i b_k^+ + \beta^* a_j^+ a_i^+ b_k) + \sum_{p,i} (\xi p_p a_i^+ + \xi^* p_p^+ a_i), \quad (4)$$

where Ω_k and $\bar{\Omega}_p$ are the frequencies associated with excitations in the heat bath and energy source respectively and λ , χ , and ξ are coupling constants for the one-quantum process, for the Fröhlich two-quantum process, and for the energy supply to the oscillating units respectively.

β is the coupling constant for the simultaneous creation and destruction two-quantum terms suggested by Lifshits. Using this hamiltonian and perturbation theory [10] yielded for the $\langle \dot{n}_i \rangle$, the average rate of change of quanta with frequency ω_i in the biological system:

$$\langle \dot{n}_i \rangle = S - \phi(T, \omega_i) [\langle n_i \rangle \exp(\hbar\omega_i/kT) - (1 + \langle n_i \rangle)] - \sum_j \Lambda(T, \omega_i, \omega_j) [\langle n_i \rangle (1 + \langle n_j \rangle) \exp(\hbar(\omega_i - \omega_j)/kT) - \langle n_j \rangle (1 + \langle n_i \rangle)] \\ - \sum_j \Gamma(T, \omega_i, \omega_j) [\langle n_i \rangle \langle n_j \rangle \exp(\hbar(\omega_i + \omega_j)/kT) - (1 + \langle n_i \rangle)(1 + \langle n_j \rangle)],$$

where

$$\phi(T, \omega_i) = \frac{4\pi|\lambda|^2}{\hbar} \sum_j N(\hbar\Omega_j) N(\hbar\omega_i + \hbar\Omega_j) \exp(\hbar\Omega_j/kT), \\ \Lambda(T, \omega_i, \omega_j) = \frac{2\pi|\chi|^2}{\hbar} \times \begin{cases} 1 + N(\hbar\omega_j - \hbar\omega_i), & \text{for } \omega_j > \omega_i, \\ N(\hbar\omega_i - \hbar\omega_j), & \text{for } \omega_j < \omega_i, \end{cases} \quad (6) \\ S = \frac{4\pi|\xi|^2}{\hbar} \sum_i \langle p_i^+ p_i \rangle_0, \quad \Gamma(T, \omega_i, \omega_j) = \frac{2\pi|\beta|^2}{\hbar} N(\hbar\omega_i + \hbar\omega_j),$$

and $N(\hbar\omega)$ is the number of excitations with energy $\hbar\omega$ in the heat bath.

If $\Lambda(T, \omega_i, \omega_j) \exp(-\hbar\omega_j/kT) = \Gamma(T, \omega_i, \omega_j)$ as Lifshits presumes, then eq. (5) can be reduced to Lifshits' form. However, as seen in eq. (6), allowing equality of $\Lambda \exp(-\hbar\omega_j/kT)$ and Γ is incorrect. Nonetheless, presupposing that the coupling constants are in the same order of magnitude for each type of two-quantum interaction, the coefficient, $\Lambda(T, \omega_i, \omega_j)$, is much larger than the coefficient, $\Gamma(T, \omega_i, \omega_j)$. This can be demonstrated readily for biological systems where ω_i is of the order of 10^{11} sec^{-1} . At room temperature $\hbar\omega_i/kT \doteq 10^{-2}$, thus Γ and Λ can be approximately written as:

$$\Gamma \doteq (2\pi|\beta|^2 kT/\hbar^2)(\omega_i + \omega_j)^{-1} \quad \text{and} \quad \Lambda \doteq (2\pi|\chi|^2 kT/\hbar)(|\omega_j - \omega_i|)^{-1}. \quad (7a, b)$$

For the Γ term, since $\omega_i \doteq \omega_j$ due to the very narrow range of normal frequencies, let $\omega_i \doteq \omega_j = \bar{\omega}$. For the Λ term, let $|\omega_j - \omega_i| = \Delta\omega$, the width of the frequency band. Thus eqs. (7) become:

$$\Gamma \doteq (\pi|\beta|^2/\hbar)(kT/\hbar\bar{\omega}) \quad \text{and} \quad \Lambda \doteq (2\pi|\chi|^2/\hbar)(kT/\hbar\Delta\omega). \quad (8a, b)$$

Since the range of normal frequencies of the oscillations is very small, $\Delta\omega \ll \bar{\omega}$. Therefore, the terms in eq. (8b) are much larger than those in eq. (8a). The simultaneous creation and destruction of quanta proposed by Lifshits can be regarded as negligible. Thus, the predominant "two-quantum" process is precisely the one postulated by

Fröhlich, and the addition of the Lifshits term does not cancel out the Bose condensation phenomenon predicted by Fröhlich.

As Fröhlich has pointed out to us, oscillations as introduced in refs. [7–9] and by the above authors form only a part of the degrees of freedom of the heat bath. Very important for the relevant energy transfer are non-oscillating systems, e.g. free ions, mobile ionic groups (dipoles), and certain low-lying electronic levels. For all of these it can be shown in a trivial way that Lifshits' contention is incorrect. In fact some of them may permit excitations in the energy range $(0, \hbar(\omega_{\max} - \omega_0))$ but not in the range $(2\hbar\omega_0, 2\hbar\omega_{\max})$, thus making two-phonon absorption or emission impossible.

References

- [1] H. Fröhlich, Phys. Lett. 26A (1968) 402.
- [2] H. Fröhlich, Int. J. Quant. Chem. 2 (1968) 641.
- [3] H. Fröhlich, Nature 228 (1970) 1093.
- [4] H. Fröhlich, Phys. Lett. 39A (1972) 153.
- [5] H. Fröhlich, Phys. Lett. 51A (1975) 21.
- [6] M.A. Lifshits, Biofiz. 17 (1972) 694.
- [7] D. Bhaumik, K. Bhaumik and B. Dutta-Roy, Phys. Lett. 56A (1976) 145.
- [8] D. Bhaumik, K. Bhaumik and B. Dutta-Roy, Phys. Lett. 59A (1976) 77.
- [9] D. Bhaumik, K. Bhaumik, B. Dutta-Roy and M. Engineer, Phys. Lett. 62A (1977) 197.
- [10] T.M. Wu and S. Austin, Phys. Lett. 64A (1977) 151.